Chapter test

30 Origins of life and the evidence for evolution

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following supports the theory of evolution?

   A. less complex fossils in lower layers of rocks.
   B. less complex fossils in upper layers of rocks
   C. the same type of fossils in upper and lower layers of rocks
   D. no fossils in any rocks ☐

2. Which of the following parts of an organism can form fossils?

   (1) bone
   (2) teeth
   (3) hair

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

3. Studying fossils shows

   (1) information about organisms which lived millions of years ago.
   (2) the age of a fossil according to the depth and layer where it was found.
   (3) how a particular group of organisms evolved.

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

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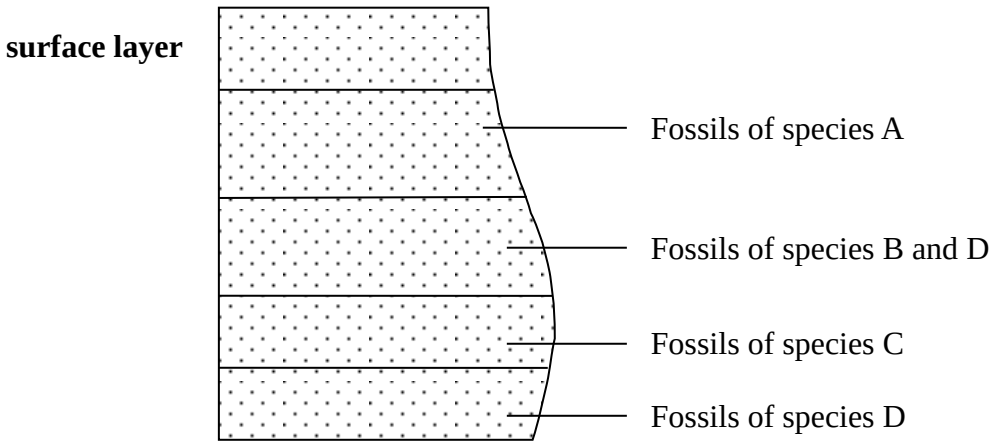
4. Human embryos and embryos of all other vertebrates have gill pouches. This suggests that

A. human embryos can breathe by gills.
B. human embryos can breathe under water.
C. human and fish are evolved from a common ancestor.
D. all vertebrates are related. ☐

5. Which of the following are homologous structures?

A. wings in birds and insects
B. fins in fish and wings in birds
C. limbs in cats and wings in bats
D. the eyes of humans and octopuses ☐

Directions: Questions 6 and 7 refer to the diagram below, which shows the fossils found in different layers of sedimentary rock.



★ 6. From the above diagram, it can be concluded that

A. species C appeared earlier than species A.
B. species D is extinct.
C. species B and D are closely related.
D. species C is evolved from species D. ☐

★ 7. Scientists studied the anatomy of the fossils of species B and C. They found that these fossils have similar limb bone structures. This suggests that species B and C

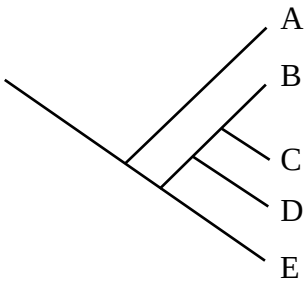
A. have highly similar DNA sequences.
B. have similar feeding habits.
C. have similar ecological niches.
D. have close evolutionary relationship. ☐

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8. Which of the following ***cannot*** be used to determine the evolutionary relationship between organisms?

   A. DNA sequence
   B. amino acid sequence
   C. bone structure
   D. the ability to inbreed ☐

Directions: Questions 9 and 10 refer to the diagram below, which shows the evolutionary tree of five species.



★ 9. Which of the following pairs of species are the most closely related?

   A. A and B
   B. B and C
   C. C and D
   D. D and E ☐

★ 10. Which of the following statements about the evolutionary history of the five species is/are ***incorrect***?

   (1) Species A is not related to the other four species.
   (2) The five species share a distant common ancestor.
   (3) Species B shares a more recent common ancestor with species C than with species D.

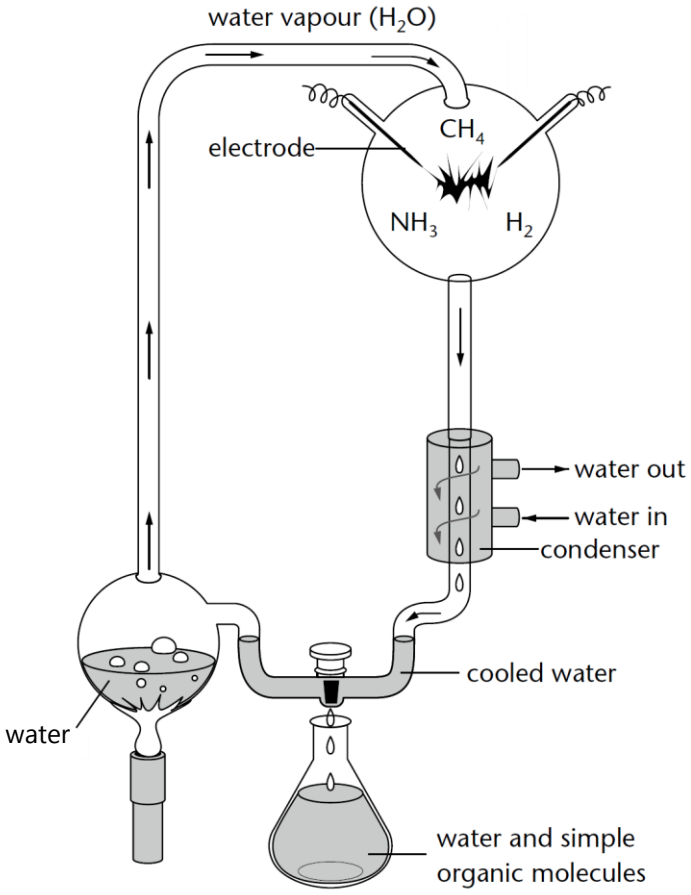
   A. (1) only
   B. (2) only
   C. (1) and (2) only
   D. (2) and (3) only ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	30.1	30.2	30.2	30.2	30.2	30.2	30.2	30.2	30.2	30.2

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B. Problem solving (22 marks)

11. The diagram below shows the set-up of the Miller-Urey experiment in 1953. The experiment simulated hypothetical conditions thought to exist on early Earth, and it tested whether organic molecules could have formed from inorganic molecules in the atmosphere of the early earth.



(a) Suggest the hypothesis of the Miller-Urey experiment. (1 mark)

(b) Based on the design of the experimental set-up, state ***two*** assumptions about the early earth made by Miller and Urey. (2 marks)

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(c) What is the purpose of passing electric sparks into the mixture of methane, ammonia, hydrogen and water vapour? (2 marks)

(d) What conclusion can be drawn from this experiment? (2 marks)

(Total: 7 marks)

12. Fossil records can serve as evidence for evolution.

(a) Describe how the remains of organisms may become preserved as fossils. (4 marks)

(b) Discuss the limitations of studying evolutionary history based on fossil records. (4 marks)

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(c) A scientist successfully isolated trace amount of DNA from a dinosaur fossil. Suggest why scientists cannot re-create dinosaurs using the isolated dinosaur DNA. (1 mark)

(Total: 9 marks)

★ 13. Red pandas are currently classified in the family Ailuridae under the mammal order Carnivora. In the past, the red panda had been classified as a relative of the giant panda, and also of the raccoon. The table below shows the scientific names of the red panda, the giant panda, and the raccoon in the 19th century and a section of DNA base sequence of a common functional gene found in these three animals.

Animal	Scientific name in the 19th century	DNA base sequence
Red panda	*Ailurus fulgens*	G A G T A C C A T T A C G C A C
Giant panda	*Ailuropoda melanoleuca*	T A G T T C C A T T A C G C T G
Raccoon	*Procyon lotor*	G A G T A C C A T T A C G C A T

(a) Based on the scientific names, which two of the animals would you expect to have a closer relationship? (1 mark)

(b) (i) Based on the DNA base sequence, which two of the animals would you suggest to be given the same genus name? (1 mark)

(ii) Explain your answer in (b)(i). (2 marks)

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(iii) Assuming the red panda, the giant panda and the raccoon have evolved from a common ancestor, construct an evolutionary tree to represent the evolutionary relationship among the three species based on analysis of DNA base sequence.

(2 marks)

(Total: 6 marks)

C. Exam challenge (8 marks)

★★ 14. Albumin is a plasma protein found in all primates, including humans, gibbons, gorillas and orangutans. Antibodies specific to human albumin are applied to the blood plasma of these primates. The antibodies will precipitate human albumin present in the blood plasma and also the albumin of closely related primates.

(a) Explain how comparing the albumin of primates could show how closely humans are related to other primates. (2 marks)

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(b) The table below shows the percentage of precipitation of albumin after mixing human albumin specific antibodies with the blood plasma of humans and other primates.

Primate	Precipitation of albumin (%)
human	100
gibbon	83
gorilla	96
orangutan	86

(i) Arrange the primates above in order of the closeness of their evolutionary relationship with humans. (1 mark)

(ii) Based on the antibody-antigen interaction, explain the results. (3 marks)

(iii) Kevin drew this conclusion based on the results:
'Orangutans are more closely related to gibbons than to humans'.
Explain whether his conclusion is correct or not. (2 marks)

(Total: 8 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	A	2.	A	3.	D	4.	C	5.	C
6.	A	7.	D	8.	D	9.	B	10.	A

(1 mark each)

B. Problem solving (22 marks)

11. (a) When water vapour, methane, ammonia and hydrogen are exposed to electric sparks, chemical reactions will occur and produce organic molecules. (1)

(b) The atmosphere of the early earth consisted of a mixture of methane, ammonia, hydrogen and water vapour. /
Lightning was frequent on the early earth. /
There was initially no organic molecules on the early earth. (Any ***two***, 2)

(c) The mixture of water vapour, methane, ammonia and hydrogen provides the raw materials for the formation of organic molecules. (1)
The electric sparks supply energy to start the reactions. (1)

(d) Organic molecules can be formed (1)
when water vapour, methane, ammonia and hydrogen reacted in the presence of electric sparks. (1)

(Total: 7 marks)

12. (a) The remains of organisms buried in mud, so that the remains would not be eaten (1)
and the rate of decomposition by bacteria was low. (1)
The mud around the hard parts of the organisms turned to rock under the pressure from the upper layers of sediment. (1)
Mineral salts deposited in the skeleton and hardened it into fossil. (1)

(b) There are missing links in the fossil record. (1)
Most fossils are incomplete or found in fragments. The entire structure of organisms has to be postulated. (1)
Not all organisms can be fossilized. The existing fossils can only represent a small portion of previous life on Earth. (1)
It is difficult to accurately determine the actual age of the fossils. (1)

(c) Most DNA is degraded. /
A complete genome of the dinosaur has to be found. /
The DNA may be contaminated. (Any ***one***, 1)

(Total: 9 marks)

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13. (a) Red panda and giant panda (1)

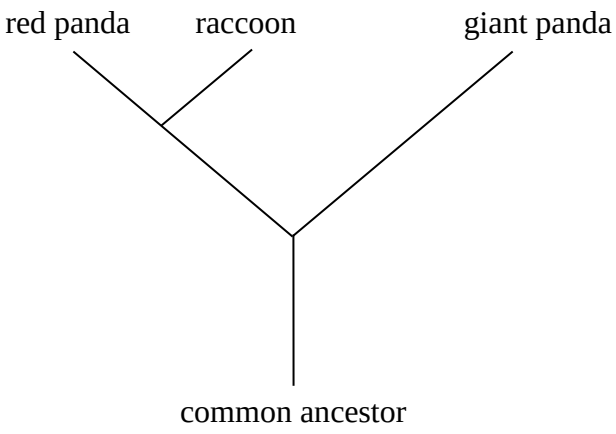
(b) (i) Red panda and raccoon (1)

(ii) There is only one difference in the DNA base sequence between red panda and raccoon, while there are 4 differences between the DNA base sequence of giant panda and both red panda and raccoon. (1)

The higher the similarity in the DNA base sequence, the more closely related the organisms. (1)

(iii) Correct tree (1)

Correct labelling (1)



(Total: 6 marks)

C. Exam challenge (8 marks)

14. (a) Albumin is made up of amino acids. Its amino acid sequence can reflect the similarities in genes / DNA coding for albumin. (1)

Fewer differences in the amino acid sequence of albumin suggest closer evolutionary relationships with other primates. (1)

(b) (i) gorilla, orangutan, gibbon (1)

(ii) Antibody has a specific antigen-binding site for human albumin. (1)

Other primates have similar albumin but with some amino acid differences. Therefore the albumins found in other primates are not a perfect match to the antibody. (1)

The larger the amount of antibody precipitated represents higher degree of similarity in albumin, which means closer relationship with humans. (1)

(iii) His conclusion is wrong. (1)

As the antibodies specific to human albumin are used in the investigation, the results only show the degree of similarity between the human albumin and primate albumins. There was no comparison between the albumin of orangutans and gibbons. (1)

(Total: 8 marks)

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